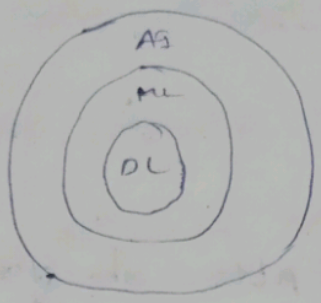
* AI:

Artificial Intelligence is a branch of computer science focusing on building systems capable of performing Task that typically require human intelligence such as learning, reasoning and problem solving Techniques

It works by processing large volumes of data through algorithm to identify patterns and make decisions

The 3 main layers of AI under the evolution of AI is considered as AI, ML & DL.



ML: It is a subset of AI where computers analyze massive data sets to learn & improve from past experiences without human intervention

DL: (Deep Learning)

An advanced ML Technique inspired by the human brain that uses multilayered Neural network to process complex, unstructured data like speeches, images, etc

Types of AI

① Weak AI / Narrow AI

systems designed to handle specific Task

Eg: Siri (Google Assistant)
self-driven cars
Game

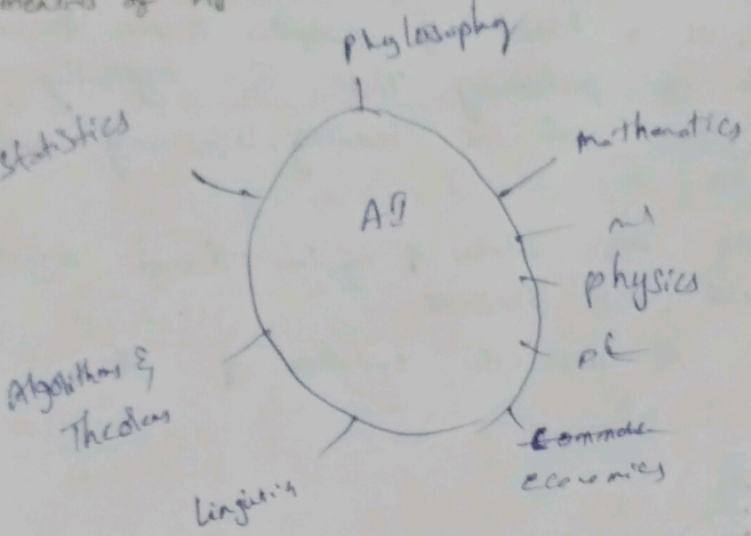
② Strong AI / General AI

which is possessing human level intelligence across all domain

→ AI can also be defined as below 4 ways

1) Systems that can think like human. Ex: chatgpt, Siri, Gemini, etc
2) Systems that can act like human Ex: Turing Test in 1956 by alan Turing
3) system that can think like Rational Ex:- laws of laws
4) System that can act rationally Ex: Rationality Agent

Fundamentals of AI



History of AI:

1943 — Warren Muccullach & Walter pitts who found first mathematical model of Artificial Neuron

1950 - Turing Test proposed by Alan Turing (proposal)

1956 — ~~Turing Test~~ word AI coined in Dartmouth Conference

1966 - ELIZA first chatbot.

1977 - 1st winter decline of AI Come into existence in business Application

1980 - Expert Systems

1987 — 2nd winter decline of AI

1997 — Deep Blue (Online chess game)

2006 — Deep learning introduced in Training Neural Networks

2012 — Computer Vision improves Image recognition

2014 — Generative Adversarial Network (GAN) Introduced

2020 → gpt-3 introduced (LLM - Large Language models)

2022 → chatgpt popularized

2023 → Rapid growth of Multimodel AI capable of understanding - AI images, Audio, Video & Code

2024-2026 → Advanced Technology with AI Agents, reasoning models, real time multimodel systems & Enterprise AI become widespread across Software development, Education, Healthcare, Science & Business sectors

State of AI:

① Large language Multimodal models:
- Modern AI models understand & Generate Text, images, Audios, Speech etc
- Agentic AI Systems: Can plan, reason, & execute complex multistep
- Advanced reasoning capability for mathematics, science, coding & Logical problem solving
- Domain specific AI models for healthcare, finances, law, education & Scientific research

② AI Powered Coding assistants for Software development, debugging & Code Generation

Real-Time multimodal interaction through speech, vision & text conversations

Vision Language Action (VLA) models

VLA models for robotics & Autonomous systems

Retrival Augmented Generation for Accurate Knowledge grounded responses

Synthetic data Generation to improve model Training by reducing dependance on real world database

Continuous learning Techniques that enable models to adapt without Forgetting

Autonomous AI Agents Capable of Colaborating with human and other AI Systems

Few more Applications of AI are

① Healthcare — disease diagnosis, medical imaging & drug discovery

② Education — personalized learning & Automated Grading

③ Finance — fraud detection, credit scoring, Algorithmic Trading

④ Banking sector - chatboard, customer support & risk Assessment

⑤ Agriculture - crop monitoring, prescription farming, pest detection

⑥ Transportation - Autonomous vehicles, Traffic Management, route Optimization

⑦ Retail sector - Product recommendation, demand forecasting, ecommerce personalized shopping experiences & virtual Assistance

⑧ Robotics - Industrial Automation, warehouse operations, NLP, language Translation Task Summarization, Sentiment Analysis

⑨ Computer vision - face recognition, Object detection, image Analysis

~~Agent & Environment:~~

AGent & Environments:

Agent: Agent is anything a entity that perseves the data in the form of input from sensors and act upon on the environment through actuators

Environment:
It is the place where Task is going to perform

Sensors:
Sensor is a form of input signals & instructions & rules & regulations by which agent can perseve the information from the environment

Actuators:
These are the output signals & actions performed by the agent on its own environment

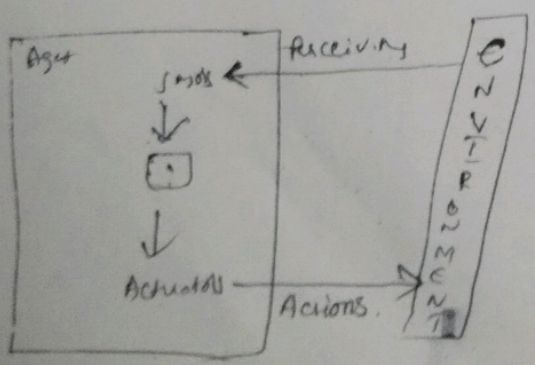
Percept sequence:
It is a previous history of agent performed actions.

Agent function:
It is a map between percept sequencing to actions

Rational Agent:
It is also called as Intelligent agent who is having ability to solve the Task in any Type of environment whether it may be static & dynamic environment and whether it may be partially observable & fully observable environment.

Good Behaviour:
A good behaviour is defined by rationality. A good agent is one that consistantly chooses actions expected to maximize it's performance measure based on it's percept sequence, environmental knowledge, and available options A few characteristics of good behavioured agent are

1. Atonomy
2. Continuous Learning
3. Adoptability
4. Proactivity
5. performance measure
6. Omni Science



Agent interacting with environment through sensors & Actuators.

* Nature of environment?

In AI, the Task environment is the external setting in which an AI agent operates, perceives, & acts.

The good behaviour & nature of the environment always defined in the four ~~methods~~ measures

P → Performance Measure

E → Environment

A → Actuators

S → Sensors.

Ex: For An Automated Taxi (A self driving car), the AI Agent must navigate a complex physical world using sensors to absorb it's surrounding and actuators to control the vehicle.

Characteristics:

The Nature of this environment is defined by several ~~different~~ distinct characteristics

① Partially observable vs fully observable: The taxi sensors like Camera, LiDAR, can't see everything: like blind corners, predict what inside a closed truck. & know exactly what other human drivers are thinking

② Multi Agent: Taxi is acting as multiagent by sharing the road with others; Pedestrians, cyclist, human drivers etc,...

③ Stochastic (Non deterministic): Outcomes of actions are not entirely Predictable. vs Static

④ Dynamic: The environment is changing continuously

⑤ Continuous vs episodic: This state (fuel, exact GPS), percept (visual Image), sensor data ~~action~~ Action (steering, braking, time are infinite.)

⑥ Unknown: vs known

Agent Type	PM	Environment	Actuators	Sensors
Taxi Driver	safe, fast, legal, Comfortable Maximize profit	Roads, other Traffic, Pedestrians, Customers	Steering, Accelerators, Brake, Signal, Horn, Display	Cameras, Sonar, Speedometer, GPS, odometer, accelerometer, engine sensors, Keyboard

PEAS → Automated Taxi Driving.

Single Agent vs Multi Agent

Stochastic vs deterministic

discrete vs Continuous

Agent Type	Performance measure	Environment	Actuators	Sensors
Medical Diagnosis System	Healthy patient Reduce cost	Patient Doctors, Staff	Tests, ECG, test Diagnosis, Treatment	Keyboard entry of symptoms, findings, patients answers
Satellite Image Analysis System	Correct Image categorization	Downlink from orbital satellite	Display of categorization	Color pixel arrays
Interactive english tutor	Student score on Test	Set of Students, Testing Agency	display of exercises, Suggestion, Corrections	Keyboard entry

* The Structure of Agents

The job of AI is to design an Agent program that implements the agent function

Agent = Architecture + program

There are multiple Agent models like

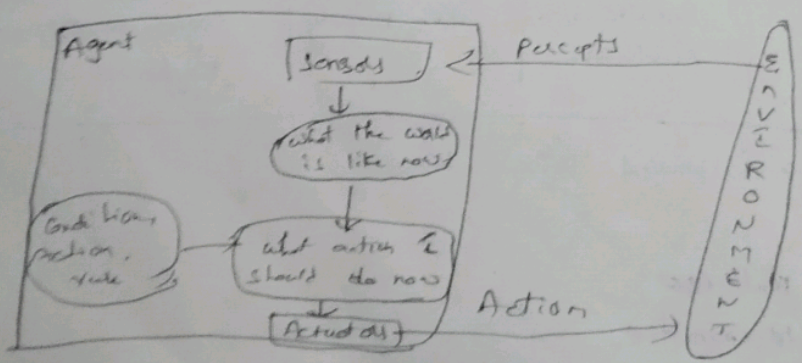
1. Simple - Reflex Agent model
2. model - Based Reflex Agents
3. Goal - Based Agents
4. utility - Based Agents
5. ~~Learning~~

Each kind of agent program combines particular components in particular ways to generate actions

① Simple - Reflex Agent model

It is the simple kind of agent. The agent select the actions on the basis of current percepts, ignoring the rest of the percept history.

for example: The Vacuum Agent is the simple Reflex agent because it's decision is based only on the current location



Algorithm:

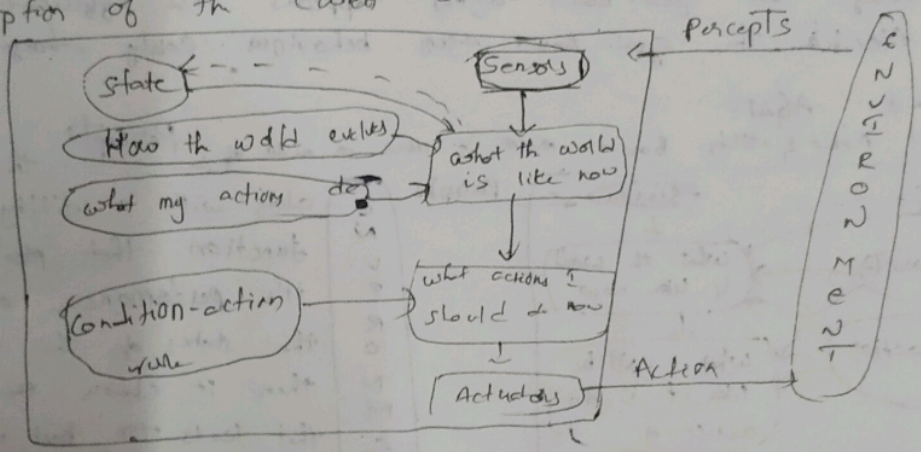
function SIMPLE-REFLEX-AGENT (percept) returns Actions
persistent : Rules, a set of condition-action rules
state ← INTERPRET-INPUT (percept)
rule ← RULE-MATCH (state, rules)
action ← rule.ACTION
return action.

2) Model-Based Reflex Agents:

Here the agent should maintain some sort of internal state that depends on the percept history and thereby reflex atleast some of the unobserved aspects of the current state.

The knowledge about the how the world works is called the model of the world. An Agent that use such model is called a model-based Agent

In the below figure the structure of the model based reflex agent with internal state showing how the current percept is combined with old internal state to generate updated description of the current state based on the Agent model

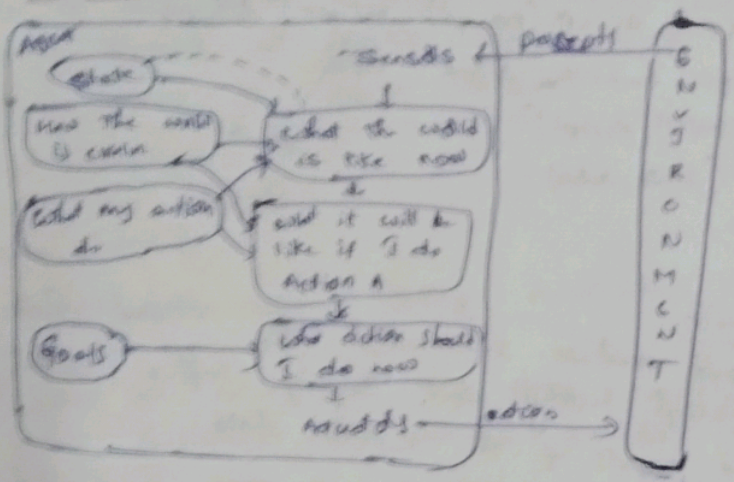


Algorithm/program:

Function MODEL-BASED-AGENT (percepts) returns action
persistent : state, the agent current world of state
model, a description of how the next state depends on current state
rule, a set of condition-action rule
action, most recent action, initally none

state ← UPDATE-STATE (state, action, percept, model)
rule ← RULE-MATCH (state, rule)
action ← rule.ACTION
return action

3) Goal-based Models:

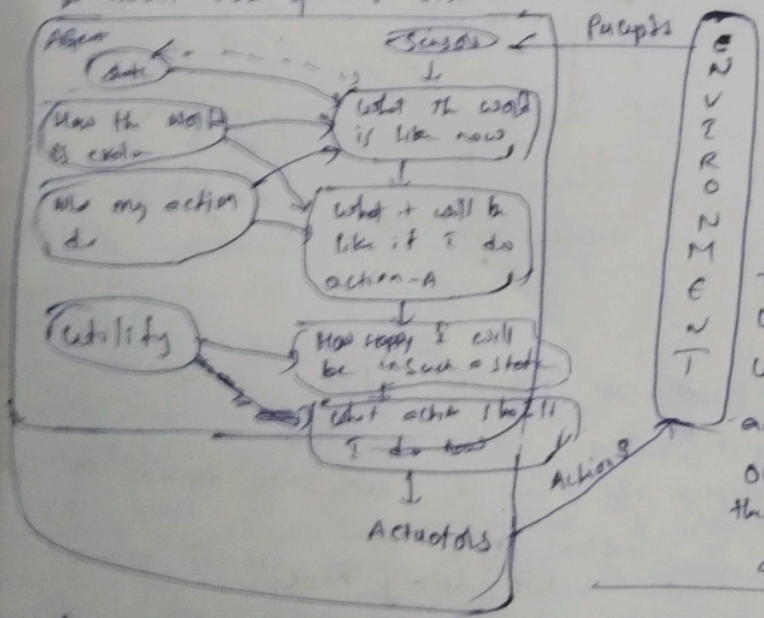


A model based & Goal-based Agent. It keep track of the world state as well as a set of goals it is trying to achieve and chooses an action that will lead to the achievements of it's goals

Knowing something about the current state of the environment is not always enough to do the Task. Sometimes the agent needs some sort of goal information that describes the situations that are desirable. Although the goal-based agent appears less efficient, it is more flexible. The goal-based agent behaviour easily changeable

4) Utility Based Agent:

A model based & utility based agent, it uses a model of the world along with a utility function that measures it's performance among the states of the world then it choose the action that leads TO best expected utility where expected utility is computed by averaging overall possible outcome states weighted by the probability of the action



Percepts

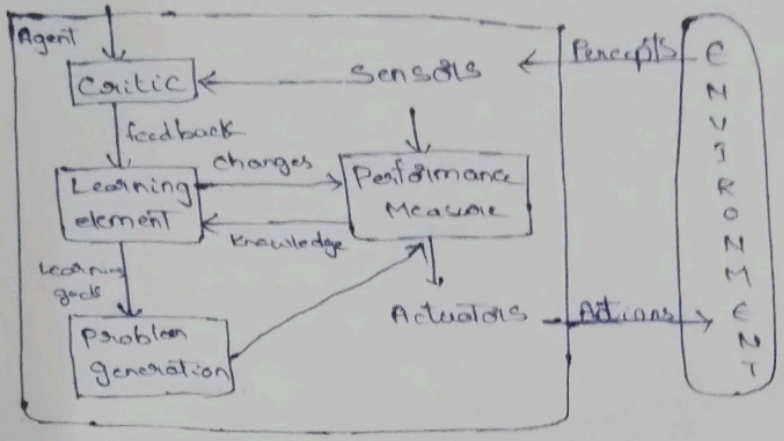
Actions

* Goals alone are not enough TO Generate high Quality Behaviour in most environments. for example: Many action sequences will get the Taxi TO it's destination thereby achieving the goal. But some are quicker, safer, more reliable or cheaper than others. So Goal just provides a distinction blw happy & unhappy states

An agent's utility function is essentially an internalization of the performance measure

Learning agent Model:

* A learning agent in AI is an autonomous system that improves its decision making over time through experience rather than relying on a fixed rules.
* By interacting with its environment it observes outcomes, adapts behaviour and optimizes performance continuously.



* A learning agent can be divided into 4 conceptual components as shown in figure. The most important distinction is in blw learning element which is responsible for making improvements and the performance element which is responsible for selecting external actions.
* The performance element is what we have previously considered to be done by the entire agent it take percepts and decide the actions.
* The learning element uses feedback from the critic and how the agent is doing & determines, how the performance element should be modified to do better in the future.
* The design of the learning element depends on the performance of the performance element.
* The critic tells the learning element how well the agent is doing with respect to a fixed performance standard.
* The last component of 'LA' is problem Generation it is responsible for suggesting actions that will lead to new & informative experiences. The problem Generations task is to suggest exploratory actions